Unit: 6 Unit 7: Factoring
Topic: Expanding Special Patterns Perfect Square Trinomial

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. What is the expansion of $(a+b)^2$?
(A) $a^2+2ab+b^2$
(B) $a^2-2ab+b^2$
(C) a^2+ab+b^2
(D) a^2-ab+b^2
(E) a^2+b^2

Q2. What is the factored form of $x^2+12x+36$?
(A) $(x+6)^2$
(B) $(x-6)^2$
(C) $(x+6)(x-6)$
(D) $(x+3)(x+12)$
(E) $(x-3)(x-12)$

Q3. A basketball player scored x points in the first half and $x+12$ points in the second half. What is the expansion of $(x+x+12)$?
(A) $2x+12$
(B) $2x-12$
(C) $x+12$
(D) $x-12$
(E) x^2+12

Q4. What is the value of a in the perfect square trinomial $a^2+20a+100$?
(A) 10
(B) 20
(C) 5
(D) 4
(E) 1

Q5. A soccer player kicks the ball x meters in the first half and $(x+5)$ meters in the second half. What is the expansion of $(x+x+5)$?
(A) $2x+5$
(B) $2x-5$
(C) $x+5$
(D) $x-5$
(E) x^2+5

Q6. What is the factored form of $x^2-16x+64$?
(A) $(x-8)^2$
(B) $(x+8)^2$
(C) $(x-8)(x+8)$
(D) $(x-4)(x-16)$
(E) $(x+4)(x+16)$

Q7. A cyclist rides x kilometers in the first hour and $(x-4)$ kilometers in the second hour. What is the expansion of $(x+x-4)$?
(A) $2x-4$
(B) $2x+4$
(C) $x-4$
(D) $x+4$
(E) x^2-4

Q8. What is the expansion of $(x-5)^2$?
(A) $x^2-10x+25$
(B) $x^2+10x+25$
(C) $x^2-10x-25$
(D) $x^2+10x-25$
(E) x^2+25

Open Ended Questions (FRQ)

Q9. A tennis player hits x aces in the first set and $(x+8)$ aces in the second set. What is the expansion of $(x+x+8)$? Show your work and provide the final answer.

Q10. Factor the perfect square trinomial $x^2+18x+81$. Show your work and provide the final answer.

Chef's Tips

- Tip 1: Emphasize the importance of factoring perfect square trinomials in real-world applications.
- Tip 2: Provide additional practice problems for students who need extra support.
- Tip 3: Consider using graphing calculators to visualize the factoring process.